CODE DIGEST
All your tech in one place

sfi
CET

ACADEMIC SUPPORT CELL
An Initiative by SFI CET Unit

CYBERSECURITY
CAREER

PHASE 01

Foundation

Technologies & Libraries
- Python Standard Library
- Linux (Ubuntu/Kali)
- Bash Shell
- Git & GitHub
- VirtualBox / VMware

Familiarize Yourself With
- Linux command line.
- File permissions.
- Bash scripting.
- SSH.
- Git workflows.
- Basic networking commands.
- Datasheets and documentation.

Goal
Build a strong foundation in programming, Linux, networking, and problem-solving.

Learn
- Computer architecture and operating system fundamentals.
- Linux fundamentals and terminal usage.
- Python programming fundamentals.
- C programming fundamentals.
- Number systems and data representation.
- Networking fundamentals.
- Study a minimum of 10–15 networking protocols (TCP, UDP, IP, ICMP, ARP, HTTP, HTTPS, DNS, DHCP, FTP, SSH, SMTP, POP3, IMAP, SNMP, etc.).
- Introduction to cryptography.

Build
- Python calculator.
- Password generator.
- File encryption utility.
- Simple TCP client/server.
- Linux automation scripts.

PHASE 02

Networking & Infrastructure Security

Technologies & Libraries
- Wireshark
- Nmap
- tcpdump
- Cisco Packet Tracer
- OpenSSL

Familiarize Yourself With
- Packet analysis.
- Port scanning.
- Network troubleshooting.
- Firewall configuration.
- Secure remote access.
- Network segmentation.

Goal
Understand how networks function and how they are secured.

Learn
- OSI and TCP/IP models.
- Routing and switching.
- VLANs.
- NAT.
- Firewalls.
- VPNs.
- DNS architecture.
- Email protocols.
- SSL/TLS.
- Wireless networking.
- Network security fundamentals.

Build
- Home lab network.
- Packet capture analysis.
- Network scanner.
- Firewall rules.
- Secure SSH server.

PHASE 03

Linux, Windows & System Administration

Learn
- Linux administration.
- Windows administration.
- User and group management.
- Process management.
- File systems.
- System services.
- Active Directory basics.
- PowerShell scripting.

Familiarize Yourself With
- User permissions.
- Service management.
- Scheduled tasks.
- System logs.
- Registry.
- Event Viewer.

Technologies & Tools
- Ubuntu
- Kali Linux
- Windows Server
- PowerShell
- Active Directory

Build
- Linux server.
- Windows Server VM.
- SSH server.
- User management scripts
- Automated backup scripts.

Goal
Develop strong system administration skills required for cybersecurity.

PHASE 04

Ethical Hacking & Penetration Testing

Technologies & Tools
- Kali Linux
- Metasploit Framework
- Burp Suite
- Gobuster
- FFUF
- Hydra
- John the Ripper
- Hashcat
- Google Dorking
- OSINT Framework

Familiarize Yourself With
- CVEs.
- Exploit databases.
- Reverse shells.
- Payloads.
- CTF platforms.

Learn
- Penetration testing methodology.
- Reconnaissance.
- Enumeration.
- Vulnerability assessment.
- Password attacks.
- Exploitation basics.
- Privilege escalation.
- Post-exploitation.

Build
- Personal penetration testing lab.
- Vulnerability reports.
- Capture The Flag writeups.
- Automated reconnaissance scripts.

Goal
Learn how attackers identify and exploit vulnerabilities in a controlled and ethical manner

PHASE 05

Web Application Security

Learn
- HTTP/HTTPS.
- REST APIs.
- Authentication.
- Sessions and cookies.
- OWASP Top 10.
- SQL Injection.
- Cross-Site Scripting (XSS).
- CSRF.
- SSRF.
- Command Injection.
- File upload vulnerabilities.

Technologies
- Burp Suite
- OWASP ZAP
- SQLMap
- Postman
- Browser Developer Tools

Familiarize Yourself With
- JWT.
- OAuth.
- Secure coding.
- API security.
- Input validation.
- Session management.

Build
- Vulnerable web application.
- Secure authentication system.
- API testing environment
- Web security assessment reports

PHASE 06

Malware Analysis, Reverse Engineering & Digital Forensics

Learn
- Linux Internals.
- Windows Internals
- Malware analysis.
- Reverse engineering.
- Memory forensics.
- Disk forensics.
- Incident response.
- Assembly language fundamentals

Familiarize Yourself With
- PE files.
- ELF binaries.
- Memory dumps.
- Registry.
- Windows API.
- Process monitoring

Technologies
- Ghidra
- IDA Free
- x64dbg
- Procmon
- Sysinternals Suite
- Volatility

Build
- Malware analysis reports.
- Memory forensic investigations.
- Reverse engineering practice
- Incident response documentation

Goal
Understand how malware operates and how digital evidence is collected and analyzed

PHASE 07

Cloud Security & DevSecOps

Learn
- Cloud architecture.
- IAM.
- Containerization.
- Kubernetes.
- Infrastructure as Code.
- CI/CD security.
- Secrets management.
- Cloud monitoring.

Familiarize Yourself With
- Docker security.
- Kubernetes security.
- Cloud logging.
- Secure deployments.
- Infrastructure hardening.

Technologies
- AWS
- Microsoft Azure
- Docker
- Kubernetes
- Terraform
- GitHub Actions

Build
- Secure cloud deployment.
- Container security lab.
- DevSecOps pipeline.
- Infrastructure automation

Goal
Secure cloud-native applications and modern deployment pipelines

PHASE 08

Blue Team Operations & Threat Detection

Learn
- Security Operations Center (SOC).
- SIEM.
- Threat hunting.
- Threat intelligence.
- Incident response.
- Log analysis.
- MITRE ATT&CK Framework.
- Detection engineering.

Technologies
- Splunk
- ELK Stack
- Wazuh
- Microsoft Sentinel
- Suricata
- Sigma Rules
- YARA

Familiarize Yourself With
- IOC analysis.
- Alert triage.
- Log correlation.
- Security monitoring.
- Threat detection workflows.

Build
- SIEM dashboards.
- Detection rules.
- Threat hunting reports.
- Security monitoring lab

Goal
Develop the ability to detect, investigate, and respond to cyber threats effectively

PHASE 09

Advanced Cybersecurity & Specialization

Learn
- Active Directory Security.
- Red Team operations.
- Purple Team methodologies.
- Advanced persistence techniques.
- Detection engineering.
- Zero Trust Architecture.
- Exploit development fundamentals
- Threat emulation.
- Enterprise security architecture

Familiarize Yourself With
- Enterprise attack chains.
- Lateral movement.
- Credential security.
- Advanced logging.
- Threat simulation.
- Security architecture reviews

Build
- Enterprise Active Directory lab.
- Red Team engagement simulation.
- Purple Team exercises.
- Threat detection rules.
- Advanced cybersecurity portfolio
- Penetration on Windows XP.

Technologies
- BloodHound
- Cobalt Strike (authorized environments only)
- CrackMapExec/NetExec
- Mimikatz
- Frida
- YARA
- Sigma
- osquery

Goal
Become an industry-ready cybersecurity professional with expertise in red teaming, blue teaming, cloud security, forensics, malware analysis, or application security.